# Albert Einstein

einstein@example.com majidfarhadi78 majid-farhadi

## Education

**University of Alberta**, Computer Engineering — Edmonton, Canada, 2023 – 2026

**Isfahan University of Technology**, Computer Engineering — Isfahan, Iran, 2018 – 2023

## Experience

**Very Large Scale Integration (VLSI) Research Lab, University of Alberta**, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

**Structured Machine Learning Research Lab, Simon Fraser University**, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

**Embedded Systems and Networks Research Lab, Isfahan University of Technology**, Undergraduate Research Assistant — Isfahan, Iran, 2021 – 2023, 2 years

- Designed an interface for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.
- Stacked on top of an IEEE 802.15.4 transceiver used in wireless sensor networks in IoT applications.
- Deployed on an Altera DE1-SoC board, featuring a Cyclone FPGA and a dual-core Cortex-A9 ARM.
- Integrated the ARM processors and FPGA fabric through a lightweight master-slave AXI bridge.
- Enabled full-duplex read/write communication between FPGA and ARM processor.
- Compiled C programs for the target ARM architecture using makefiles.
- Accessed the FPGA from C programs via virtual addresses provided by memory mappings.
- Offloaded computation from the processor and exploited the FPGA's high throughput.

**Kian-Pardaz Naghsh-e-Jahan Engineering Company**, Intern — Isfahan, Iran, 2021 – 2021, 1 year

- Designed a PCB and implemented a web client for a smart thermostat on an ESP8266-12E module.

## Volunteer

**People's Climate March**, Lead Organizer — Zurich, Switzerland, Apr 2014 – July 2015

Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.

- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

## Awards

**Nobel Prize in Physics** Nov 1921
The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'
Royal Swedish Academy of Sciences
www.nobelprize.org/prizes/physics/1921/einstein/biographical

**Max Planck Medal** 2029
Awarded for outstanding scientific achievement
German Physical Society

## Publications

**Zur Elektrodynamik bewegter Körper**
It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

**Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt**
In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.
Albert Einstein
de.wikisource.org/wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspur

**Die Grundlage der allgemeinen Relativitätstheorie**
The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).
Albert Einstein
de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

## Skills

**Physics**

## Languages

**German**
Native speaker

**English**
Fluent

## Interests

**Physics**

## Certificates

**Machine Learning** Jan 2018

**Quantum Computing** Jan 2018

**Quantum Information** Jan 2018

## Projects

**Quantum Computing** Jan 2018 – Jan 2018

Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

## References

**Professor John Doe**

**Professor Jane Smith**